MILESTONE 1 · FORMULA 1 PODIUM PREDICTION

Formula 1 Podium Prediction

Milestone 1 analytical report: data cleaning, data-engineering questions, feature engineering, modeling, experiments and explainability.

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Kaggle notebook: https://www.kaggle.com/code/seifezz11/f1-project-ms1
GitHub repository: https://github.com/YShimy/f1-project-ms1
Dataset: Formula 1 World Championship 1950–2024 (Kaggle, rohanrao)
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1. Task and setup

We predict whether a driver will finish on the podium (positionOrder ≤ 3) in a race, using only information available after qualifying and before the race starts. The unit of analysis is one driver entering one race (raceId, driverId). The positive class is rare: about 15% of rows in the modeled seasons.

Item	Definition
Target	podium = 1 if positionOrder ≤ 3, else 0 (positionOrder is filled for every row, unlike position)
Prediction cutoff	After qualifying, before the race. Current-race results, status, points, laps, pit stops, lap times and post-race standings are excluded
Modeled seasons	2003–2024 (complete qualifying data, Section 2.6); earlier seasons are used only as history for form features
Split	By season: train 2003–2018, validation 2019–2021, test 2022–2024
Primary / secondary metric	PR-AUC / ROC-AUC (Section 5.2)

2. Data audit and cleaning

Raw CSV files were loaded into a separate dictionary and never modified; every cleaning step works on copies, which also allows a before/after comparison.

2.1 EDA before cleaning and sentinel handling

The raw results table reports no missing values, but numeric columns such as position and milliseconds are typed as text because missing values are stored as the literal string \N. Six tables contain this sentinel.

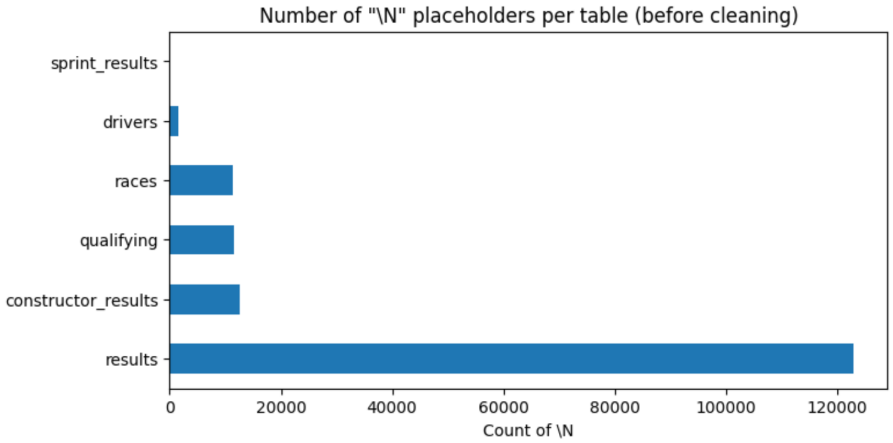


Figure 1. Count of \N placeholders per table before cleaning (8 tables have none).

Sentinel handling. All 14 tables were reloaded with `pd.read_csv(..., na_values='\\N')`, which converts every \N into a real missing value (NaN). After this step no \N remains in any table, and pandas can see the missing values: results now shows 122,887 NaN (position: 10,953, i.e. 41%; time and milliseconds: 19,079, 71%; fastest-lap fields: 18,507, about 69%), and columns such as position, milliseconds and rank became numeric (float64). Qualifying shows 11,668 NaN instead of 11,600 \N: the extra 68 were empty cells already in the file.

2.2 Semantic typing

Race dates (races.date) and driver dates of birth (drivers.dob) were converted from text to datetime with `pd.to_datetime`, which is needed to sort races in time order. Qualifying lap times written as minutes:seconds ("1:27.452") were converted to seconds (1 × 60 + 27.452 = 87.452) by a small parser; a missing or malformed time returns NaN instead of an invented value. The converted values were stored in new columns q1_sec, q2_sec and q3_sec next to the originals, so the conversion can be checked: the number of missing values is identical before and after (q1 156, q2 4,647, q3 6,865), so no time was lost or broken.

2.3 Primary and foreign keys

For each of the 14 tables we counted rows with an empty primary key and rows repeating a key. lap_times and pit_stops have no single ID, so their composite keys (raceId, driverId, lap) and (raceId, driverId, stop) were checked. Result: 0 empty and 0 duplicated keys in every table. For 22 foreign-key relationships (for example results.driverId → drivers.driverId, races.circuitId → circuits.circuitId, races.year → seasons.year) we counted values that do not exist in the referenced table: 0 broken links in all 22. No change to the data was needed; every join used later is safe.

2.4 Grain: one row per (raceId, driverId)

85 driver-race cases (176 rows) have more than one row, almost all from 1950–1958 (shared drives).

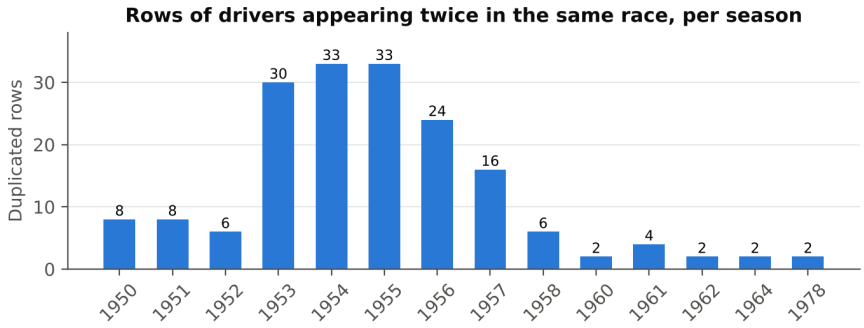


Figure 2. Duplicated rows per season.

Rule. We keep the car the driver started in, identified only from grid position (pre-race information). In 82 of the 85 cases all rows had a real grid position, because a car taken over mid-race keeps the grid slot of the driver who started it. Since each grid slot belongs to one car, the driver's own starting car is the row whose grid slot no other driver in that race holds; this resolves 53 cases. For the other 32 (31 with no such row, 1 with two), the data cannot tell which car he started in, so one row is chosen at random with a fixed seed (random_state = 42, reproducible). **Why:** choosing the best result would select rows using post-race information and give these drivers two chances at a podium; choosing the better grid would bias toward favourable starts. 91 rows were removed (26,759 → 26,668), leaving 0 duplicates.

2.5 Missingness

- **grid = 0**: of 1,638 rows, 1,566 (96%) completed 0 laps, mostly "Did not qualify", "Did not prequalify" or "Withdrew"; the other 72 drivers did race, most likely from the pit lane. So grid 0 is not a starting slot and is handled separately in feature engineering (Section 4.3).
- **position**: 10,920 of 10,922 missing values are explained by a letter code in positionText (R, F, W, N, D, E). Two rows (race 71, drivers 18 and 11) held a number ("11" and "14"); after confirming no other driver in that race held the same position, position was filled from positionText. This column is not used by the target (positionOrder).
- **Columns used for joins and the questions** (IDs, grid, positionOrder, status, race date/year/round, nationality, circuit country, constructor name): 0 missing.

2.6 Qualifying coverage and scope

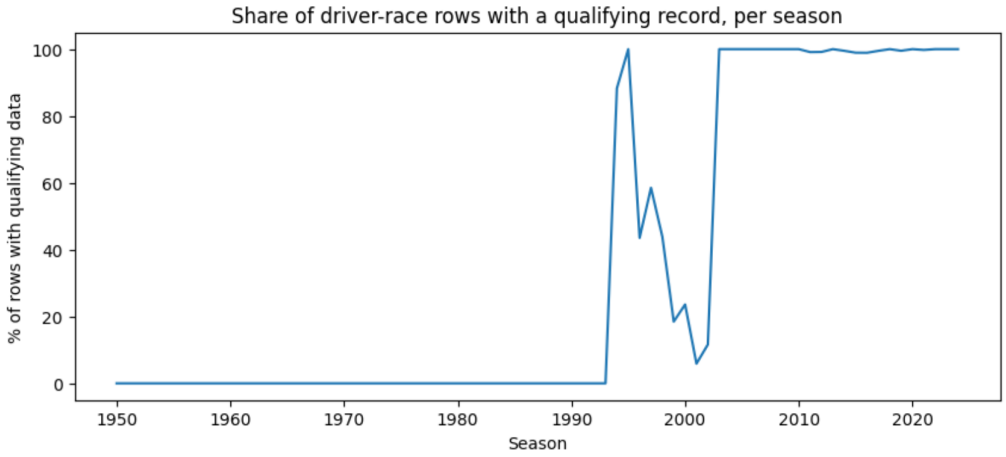


Figure 3. Share of driver-race rows with a qualifying record, per season.

There is no qualifying data before 1994, coverage is patchy in 1996–2002 (down to about 5%), and close to 100% from 2003. To avoid inventing qualifying values, the modeling table uses 2003–2024, removing whole seasons rather than individual drivers. Within 2003–2024 (8,970 rows), we removed the 36 rows whose status is a non-start known before the race (rule: status in Did not qualify, Did not prequalify, Withdrew, Injury, 107% Rule; it never uses the race result) (Withdrew 31, of which 14 at the 2005 United States Grand Prix; Did not qualify 4; Injury 1). Pit-lane starters and cars that failed at the start are kept, since they were real starters at prediction time.

2.7 EDA after cleaning

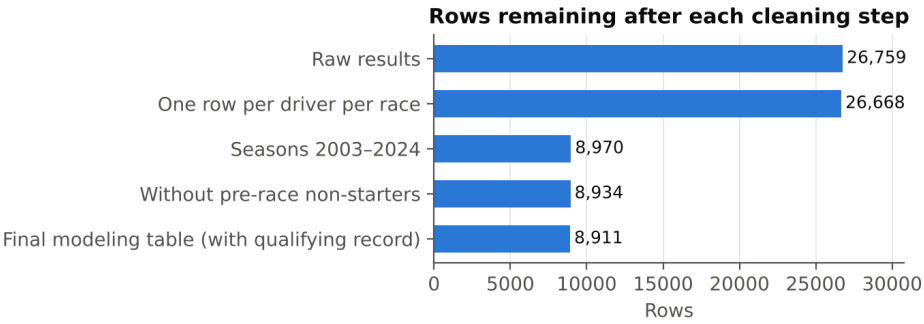


Figure 4. Rows after each cleaning step. The largest drop is the season restriction.

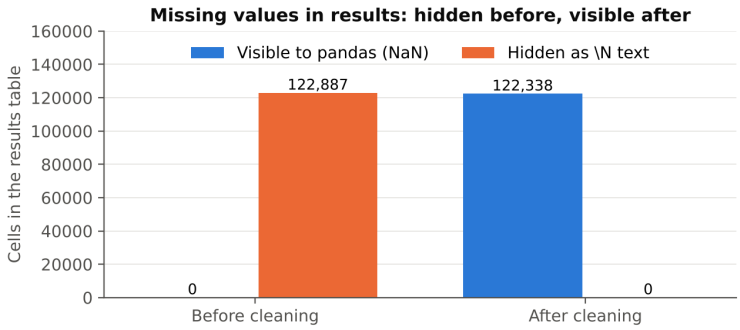


Figure 5. Missing values in results before (hidden as \N) and after cleaning (visible).

3. Data-engineering questions

3.1 Q1: front-row conversion by circuit, before and after 2014

Method. Joins: results → races → circuits (circuit name) and results → qualifying (qualifying position). A front-row start is grid 1–2; conversion = share of front-row starters finishing on the podium, per circuit, for 2003–2013 and 2014–2024 (11 seasons each). Only circuits with at least 10 front-row starts in each era are ranked (14).

Circuit	Starts 03–13	Rate 03–13	Starts 14–24	Rate 14–24	Rank before	Rank after
Suzuka Circuit	18	72%	18	89%	4	1
Yas Marina Circuit	10	60%	22	86%	10	2
Circuit de Monaco	22	77%	20	85%	1	3
Circuit de Spa-Francorchamps	18	72%	22	82%	4	4
Circuit de Barcelona-Catalunya	22	68%	22	82%	6	4
Autódromo José Carlos Pace	22	59%	19	79%	12	6
Shanghai International Circuit	20	65%	14	79%	8	7
Circuit Gilles Villeneuve	20	60%	18	78%	10	8
Silverstone Circuit	22	73%	24	75%	3	9
Autodromo Nazionale di Monza	22	77%	22	73%	1	10
Marina Bay Street Circuit	12	58%	18	72%	14	11
Albert Park Grand Prix Circuit	22	59%	18	72%	12	11
Bahrain International Circuit	18	67%	24	67%	7	13
Hungaroring	22	64%	22	59%	9	14

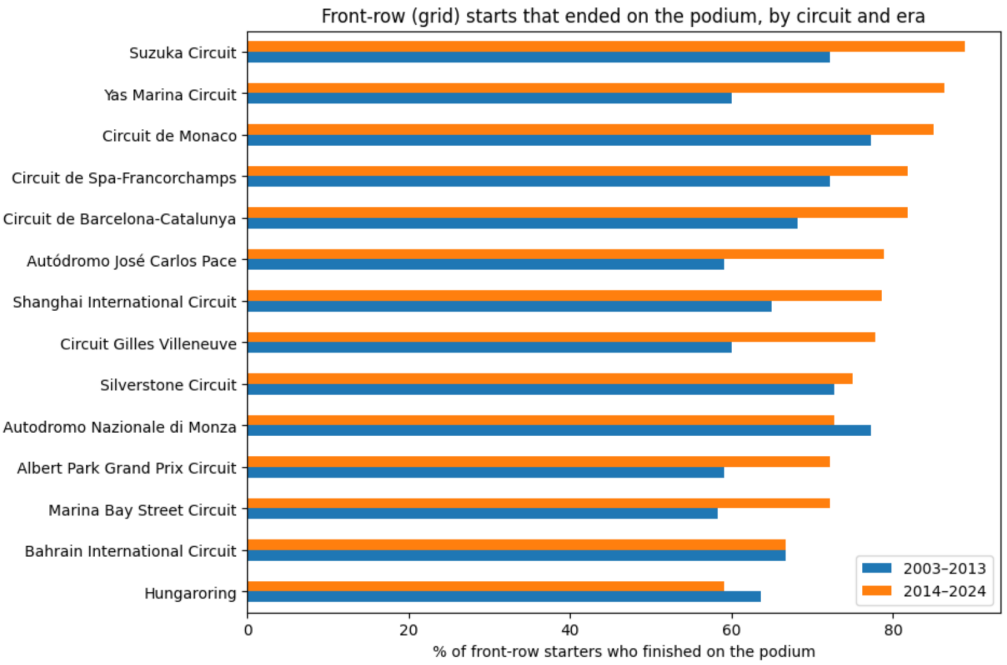


Figure 6. Front-row (grid) conversion rate by circuit and era.

Qualifying vs. grid. Of 854 front-row qualifiers, 38 did not start on the front row (penalties) and 36 others were promoted to it: the definitions differ in about 4% of front-row slots. With the qualifying definition, most rates are similar, but some ranks change (Monaco before 2014: 1st by grid, 6th by qualifying; Interlagos after 2014: 6th vs. 13th).

Front-row group (2003–2024)	Drivers	Podium rate
Qualified and started on the front row	816	73%
Qualified on the front row, started further back (penalty)	38	29%
Started on the front row without qualifying there	36	64%

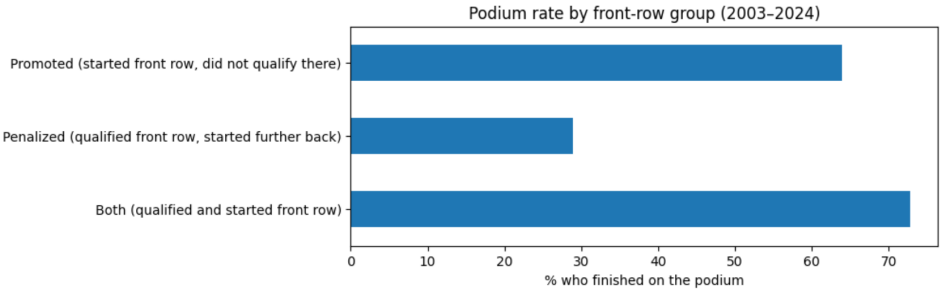


Figure 7. Podium rate by front-row group.

ANSWER

From 2014 on, Suzuka (89%), Yas Marina (86%) and Monaco (85%) convert most often. The ranking changed: conversion rose at almost every circuit (mostly 58–77% before, 72–89% after); Yas Marina rose from 10th to 2nd, Monza fell from 1st to 10th, Silverstone from 3rd to 9th; Monaco stayed near the top.

The actual starting position matters more than qualifying speed alone (73% vs. 29%).

Grain, missingness, cutoff. One row per driver and race; 2003–2024 because qualifying is complete only from then; pre-race non-starters excluded. Grid and qualifying are pre-race; podium is the outcome. With 18–24 starts per circuit and era, one podium moves a rate by about 5 points, so only large rank changes are meaningful. Penalties often come from engine changes, so the 29% is not purely a starting-position effect.

3.2 Q2: home advantage after controlling for starting position

Method. Joins: results → drivers (nationality) and results → races → circuits (country). Nationalities were mapped to country names (for example British → UK), after stripping whitespace ("Argentinian ") and unifying "USA" and "United States". 365 of 8,934 rows (4%) are home races. Without control: home 14.2%, away 14.4%. We then compared within grid groups (1–3, 4–6, 7–10, 11+; pit-lane starts counted in 11+), and, as a stricter check, compared each driver with his own away results from the same grid group (expected home podiums = his away rate × his home starts).

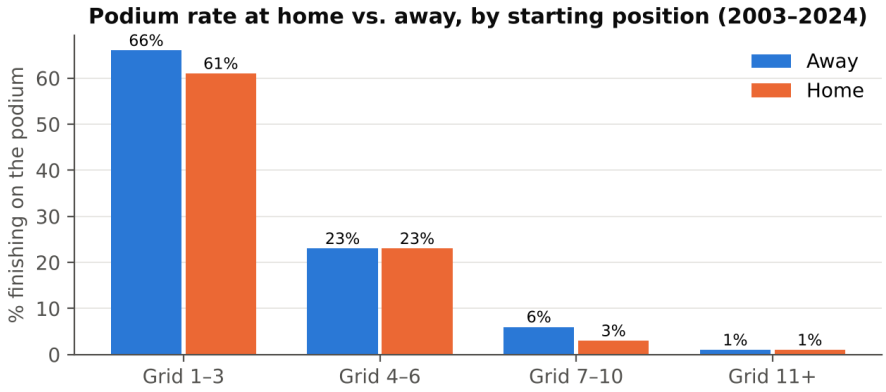


Figure 8. Podium rate at home and away by grid group.

Grid group	Home starts	Actual home podiums	Expected from own away form
1–3	61	37	40.5
4–6	47	11	11.0
7–10	76	2	3.5
11+	179	2	2.0
All	363	52	56.9

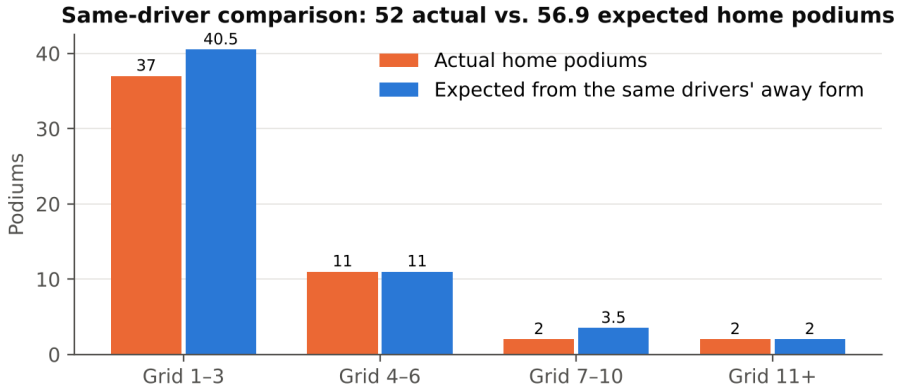


Figure 9. Same-driver comparison of actual and expected home podiums.

ANSWER

No home advantage in 2003–2024. Within grid groups, home and away rates are equal (4–6: 23%; 11+: 1%) or slightly lower at home (1–3: 61% vs. 66%; 7–10: 3% vs. 6%), differences of 2–3 podiums. Comparing each driver with himself (363 of 365 home races matched), drivers scored 52 home podiums against about 57 expected, within normal random variation (about ±7).

Grain, missingness, cutoff. One row per driver and race (2003–2024); no nationality or country missing; grid is pre-race. Limitation: nationality does not always match where a driver lives; home races are rare, so home rates are imprecise.

3.3 Q3: mechanical-retirement rate by constructor

Method. Joins: results → status, constructors, races. Rate = mechanical retirements / race starts, per constructor and era (2014–2021, 2022–2024). Pre-race non-starters are excluded since they are not starts. A status counts as mechanical only if it names a failed car part (Engine, Power Unit, Gearbox, Brakes, Hydraulics, Electrical, Suspension, Overheating, fuel and

cooling systems, etc.). Crashes, driver causes, rule decisions, operational causes (Wheel, Wheel nut, wings, Undertray, Puncture, Tyre, Out of fuel) and "Retired" (no reason given, 50 rows) are not counted. Every team has at least 31 starts (about one season), so all are shown with their number of starts.

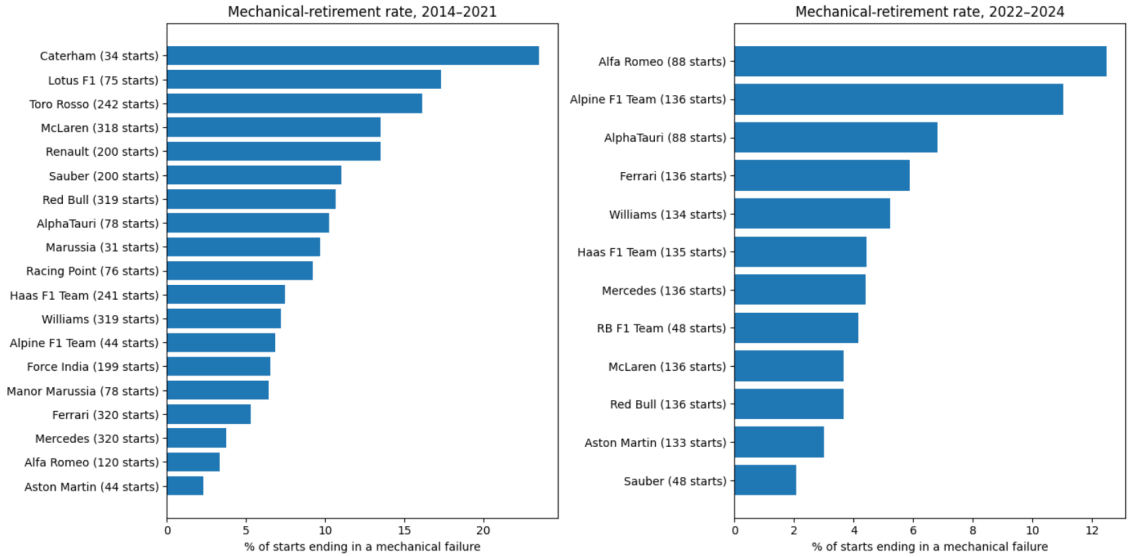


Figure 10. Mechanical-retirement rate by constructor and era (labels show starts).

2014–2021 (highest)	Rate	2022–2024 (highest)	Rate
Caterham (34 starts)	23.5%	Alfa Romeo (88)	12.5%
Lotus F1 (75)	17.3%	Alpine F1 Team (136)	11.0%
Toro Rosso (242)	16.1%	AlphaTauri (88)	6.8%
McLaren (318)	13.5%	Ferrari (136)	5.9%
Renault (200)	13.5%	Williams (134)	5.2%

ANSWER

2014–2021: Caterham (small sample), Lotus F1, Toro Rosso and McLaren had the highest rates; Mercedes (3.8%) and Ferrari (5.3%) were the most reliable full-time teams. 2022–2024: Alfa Romeo and Alpine highest; Red Bull and McLaren 3.7%.

Overall reliability improved (about 9% of starts with a part failure in 2014–2021, 300 of 3,258, vs. about 6%, 76 of 1,354). McLaren moved from one of the least to one of the most reliable teams.

Grain, missingness, cutoff. One row per race start. Renamed teams are counted separately (each name has its own ID). Q3 is descriptive and uses status, a post-race field, by design; status is never a model feature.

4. Features and data preparation

4.1 Feature evidence from the analysis

- Q1: starting position is the strongest pre-race signal, and qualifying speed adds information → grid and qualifying features.
- Q2: no home advantage → no home feature.
- Q3: reliability varies by team and era → recent team reliability.
- The front row became more decisive after 2014 → recent form and a time-based split.

4.2 Selected features and validity at the pre-race cutoff

All features are computed on the full race history (1950–2024, without pre-race non-starters), ordered by date, before restricting to 2003–2024, so early 2003 rows have proper history. Form features use shift(1) before a rolling window, so the current race is never included.

Feature	Definition	Why valid before the race
grid	Starting position; pit-lane starts (grid 0) set to one place behind the last car	Fixed before the start
pit_lane_start	1 if the driver started from the pit lane	Known before the start
quali_pos	Qualifying position	Saturday qualifying
q1_gap_pct	Gap to the fastest Q1 time, % (Q1 is the only session all drivers run under the same conditions)	Saturday qualifying
driver_podiums_last5	Podiums in the driver's previous 5 starts	Earlier races only (shift 1)
driver_avg_finish_last5	Mean finishing position over previous 5 starts	Earlier races only
driver_prev_races	Number of previous starts	Earlier races only
days_since_last_race	Days since the driver's previous start (shows when form is old)	Calendar
team_podiums_last5	Team podiums (both cars) in its previous 5 races	Earlier races only
team_mech_rate_last10	Share of the team's starts with a mechanical failure in its previous 10 races (10, not 5, because failures are rare: about 6–9% of starts)	Earlier races only
driver_champ_rank_before	Championship rank from the previous race's standings	Standings shifted by one race
team_champ_rank_before	Team championship rank from the previous race	Standings shifted by one race
4 flags	team_form_missing, driver_rank_missing, team_rank_missing, q1_gap_missing	Derived from the above

Design choices: championship rank rather than points (the points system changed in 2010); team history restarts after more than four years without racing (7 comebacks in 2003–2024, e.g. Mercedes 2010 after 1955, Aston Martin 2021), while driver history is kept and days_since_last_race added; no sprint feature (it does not exist in training seasons).

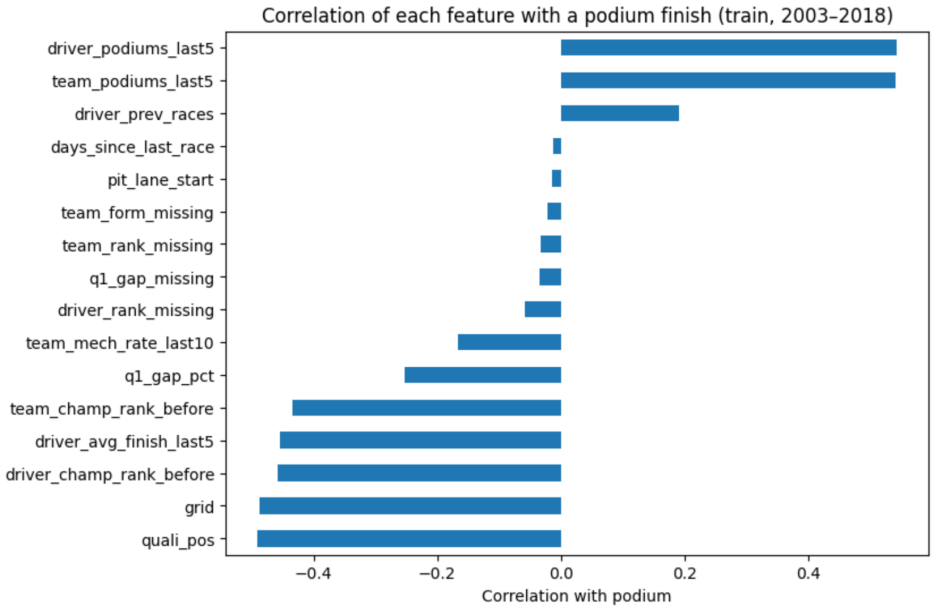


Figure 11. Correlation of each feature with the podium on the training seasons.

Leakage checks. No post-race column is in the feature list; no feature has a correlation close to ±1; for Fernando Alonso (2012, round 10), podiums (4) and average finish (2.6) recomputed by hand from his five previous starts match the table. For Lewis Hamilton's first 2020 race, the rank before the race is 1st (his final 2019 rank) while the rank after that race is 4th, confirming the standings shift.

4.3 Missing values and preprocessing

23 rows without a qualifying record were removed (no substitute values). Remaining missing values come from rookies, new teams and missing Q1 times: previous podiums are set to 0 (a true value), other features are filled with the median of the training seasons only, and a flag marks every filled value. Removing all rows with any missing value would drop 330 rows (3.7%; train 4.2%, validation 3.0%, test 1.8%), mostly rookies, first races of new teams and drivers without a Q1 time, which the model would then never learn about.

Feature	Missing	Cause	Filled with
driver_podiums_last5 / team_podiums_last5	89 / 55	First race of a driver / team	0 (true value)
driver_avg_finish_last5	89	Rookie's first race	11.4 (training median)
days_since_last_race	89	Rookie's first race	14 days
team_mech_rate_last10	55	Team's first race	0.10
driver_champ_rank_before	168	No previous standings	11 (+ driver_rank_missing = 1)
team_champ_rank_before	79	New or returning team	6 (+ team_rank_missing = 1)
q1_gap_pct	124	No Q1 time set	1.36% (+ q1_gap_missing = 1)

Rookies are flagged by driver_prev_races = 0 and new teams by team_form_missing = 1. Grid 0 (74 pit-lane starts in the final table) was replaced by one place behind the last car of that race, with pit_lane_start = 1, because as a number 0 would rank ahead of pole position. Final table: 8,911 rows, 16 features, 0 missing values. Features are then standardised (mean 0, standard deviation 1 on the training seasons) with a scaler fitted on training data only and applied

unchanged to validation and test; this puts features with very different ranges (championship ranks are small integers, while days since the last race range from about 7 to over 1,200 for comebacks) on a common scale for logistic regression and the neural network. The random forest uses unscaled values, since its splits do not depend on scale.

4.4 Split

Part	Seasons	Rows	Podium rate
Train	2003–2018	6,362	14.1%
Validation	2019–2021	1,195	15.1%
Test	2022–2024	1,354	15.1%

A season-based split mirrors real use (predicting future races from past ones). 2019 is in validation rather than training to give about 180 validation podiums instead of 117, making threshold and model selection more stable. Test (also a new regulation era) is used once, at the end.

5. Models and evaluation

5.1 Models

Model	How it works	Settings (chosen on validation)	Limitations
Logistic regression (baseline)	Weighted sum of features → sigmoid probability	max_iter 1000, scaled features	Only straight-line effects; one weight per feature for everyone
Random forest	300 decision trees on random rows/features, averaged; captures feature combinations	max_depth 4 (from 4, 6, 8, 12), min_samples_leaf 5	Deeper trees overfit; less interpretable
Shallow neural network	16 → Dense 32 ReLU → Dense 16 ReLU → 1 sigmoid; Adam, binary cross-entropy	Early stopping on validation PR-AUC (patience 15); stopped at epoch 32, weights restored from the best epoch (about 17); batch 32	Needs more data; tuned on validation seasons

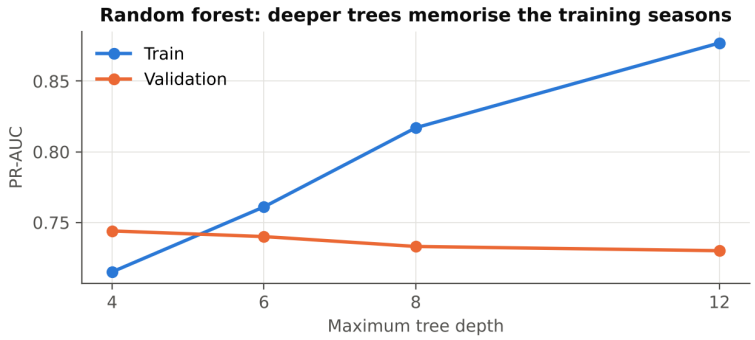


Figure 12. Random-forest depth selection: deeper trees overfit the training seasons.

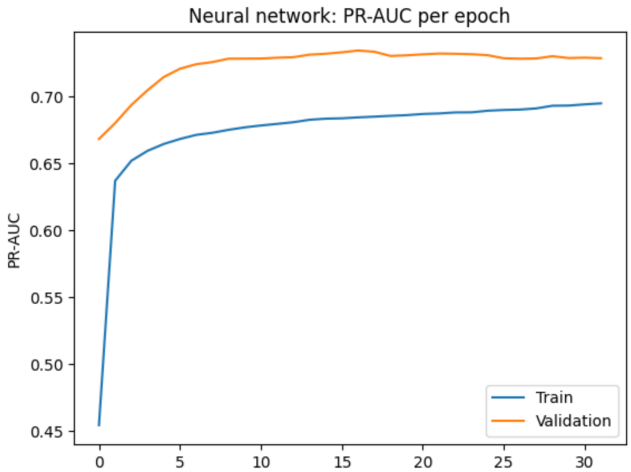


Figure 13. Neural-network PR-AUC per epoch (training and validation).

5.2 Metrics and threshold

Predicting "no podium" for everyone is 84.9% accurate on validation, so accuracy is not used. **Primary metric: PR-AUC**, because podiums are rare and the precision-recall curve focuses on finding them (random model ≈ 0.15). **Secondary: ROC-AUC** (overall separation, random = 0.5). F1, precision and recall are reported at each model's probability threshold, chosen as the

best validation F1 over 0.05–0.94, never on test.

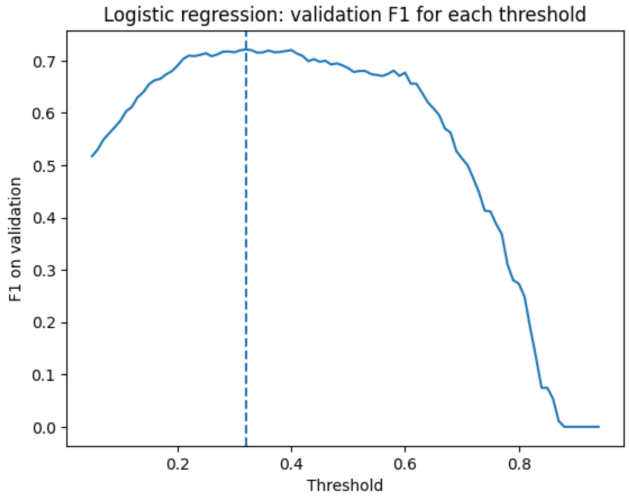


Figure 14. Validation F1 by threshold for logistic regression (chosen 0.32; flat optimum between about 0.25 and 0.45).

5.3 Results

Model	Part	Thr.	ROC-AUC	PR-AUC	F1	Precision	Recall
Logistic regression	train	0.32	0.928	0.682	0.663	0.594	0.751
	validation	0.32	0.938	0.731	0.721	0.653	0.806
	test	0.32	0.935	0.706	0.674	0.612	0.750
Random forest (depth 4)	train	0.31	0.938	0.715	0.686	0.614	0.778
	validation	0.31	0.937	0.744	0.727	0.662	0.806
	test	0.31	**0.940**	**0.721**	**0.691**	0.615	0.789
Neural network (32-16)	train	0.40	0.937	0.700	0.667	0.624	0.717
	validation	0.40	0.934	0.734	0.735	0.684	0.794
	test	0.40	0.932	0.685	0.664	0.638	0.691

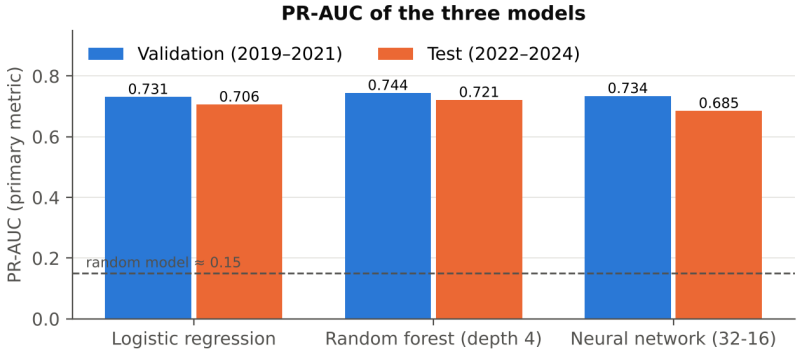


Figure 15. PR-AUC on validation and test.

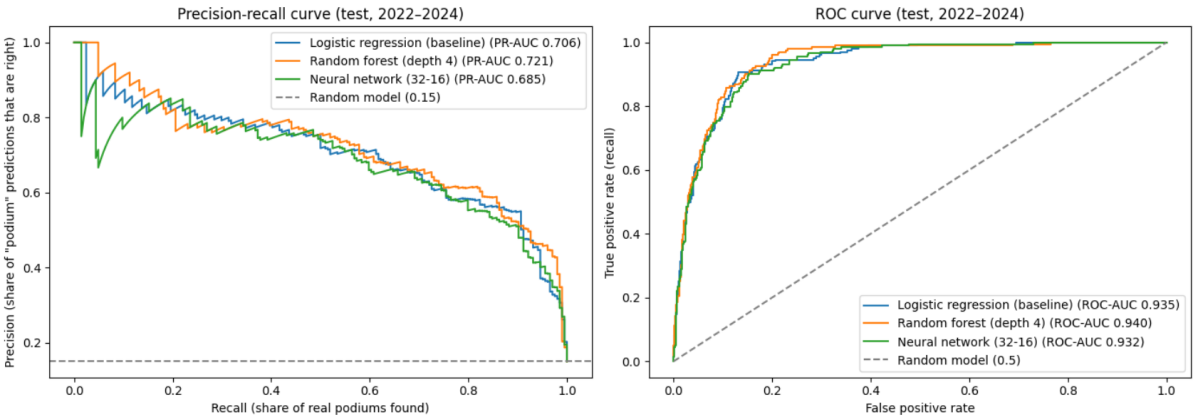


Figure 16. Precision-recall and ROC curves on the test seasons.

Training and validation scores are close for all models, so none overfits. The random forest is best on every test metric (ROC-AUC 0.940, PR-AUC 0.721, F1 0.691), consistent with its best validation PR-AUC, and is our final model. All models lose 0.02–0.05 PR-AUC on test, which coincides with the 2022 regulation change; the network loses most, plausibly because both early stopping and its threshold were tuned on 2019–2021. The neural network does not beat the baseline: the main relationships are monotonic and already summarised by the engineered features. The small gap between models suggests features matter more than model choice.

6. Experiments

6.1 Feature-group ablation (random forest, validation)

Removed group	PR-AUC	ROC-AUC	PR-AUC drop
None	0.744	0.937	—
Start (grid, pit_lane_start)	0.706	0.930	0.038
Qualifying	0.734	0.935	0.009
Driver form	0.741	0.937	0.003
Championship	0.741	0.936	0.003
Team form	0.746	0.938	−0.002

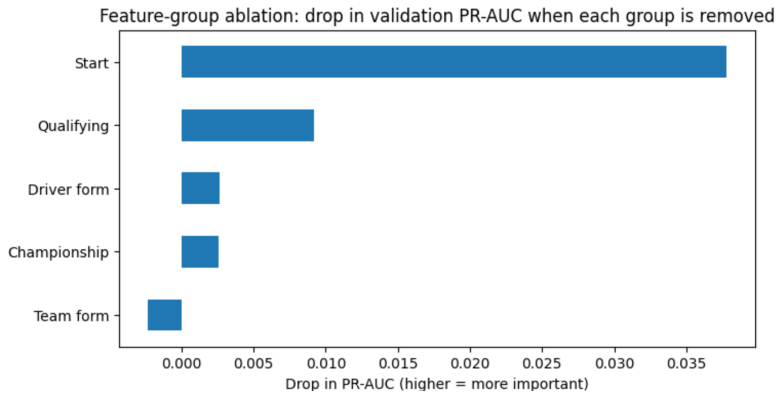


Figure 17. Drop in validation PR-AUC when each feature group is removed.

The start group matters most even though qualifying position remains, so the actual grid (after penalties) carries information beyond qualifying, consistent with Q1. Driver form, team form and championship rank each matter little alone because they are redundant (they all describe recent strength of driver and car), not because they are uninformative.

6.2 Preprocessing-order experiments (logistic regression)

Version	PR-AUC val	PR-AUC test
Medians and scaler fitted on training seasons only (ours)	0.7315	0.706
Experiment A: filling medians fitted on all seasons	0.7313	0.706
Experiment B: scaler fitted on all seasons	0.7314	0.706

Fitting preprocessing on all seasons (before splitting) uses validation and test information. Here it changes PR-AUC by at most 0.0002, because only 3.7% of rows needed filling and feature distributions are stable across seasons. Our results therefore do not rely on leakage; we keep the train-only order as the correct procedure.

Further comparisons documented in the notebook: best-of-Q1/Q2/Q3 time vs. Q1-only gap; default threshold 0.5 vs. tuned (F1 0.686 → 0.721 for the baseline); four tree depths; three model families.

7. Explainability

INTERPRETATION

These explanations describe what the model relies on to make predictions. They are not causal effects: a feature being important does not mean it causes podium finishes.

7.1 Global: permutation importance

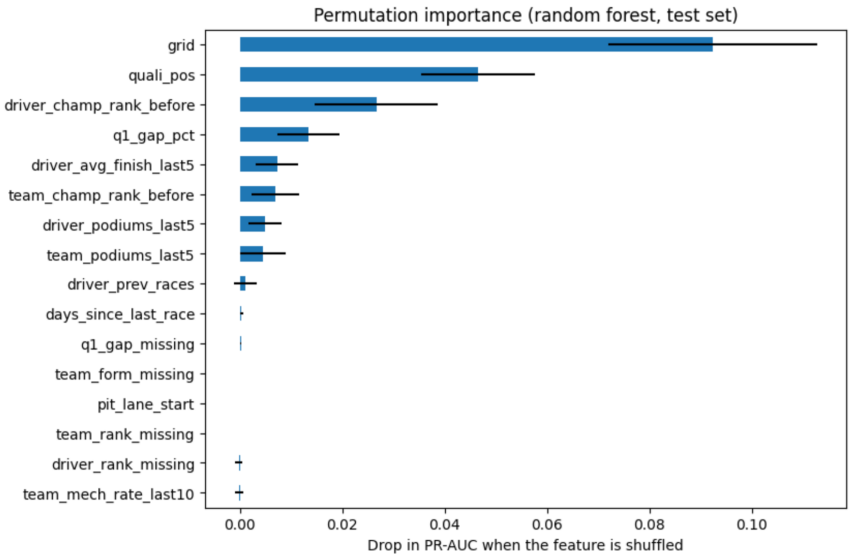


Figure 18. Permutation importance (drop in test PR-AUC when a feature is shuffled, 10 repeats).

grid (0.092), quali_pos (0.047), driver_champ_rank_before (0.027) and q1_gap_pct (0.013) dominate; flags, pit-lane starts and team reliability contribute about 0. This agrees with the ablation.

7.2 Global: SHAP

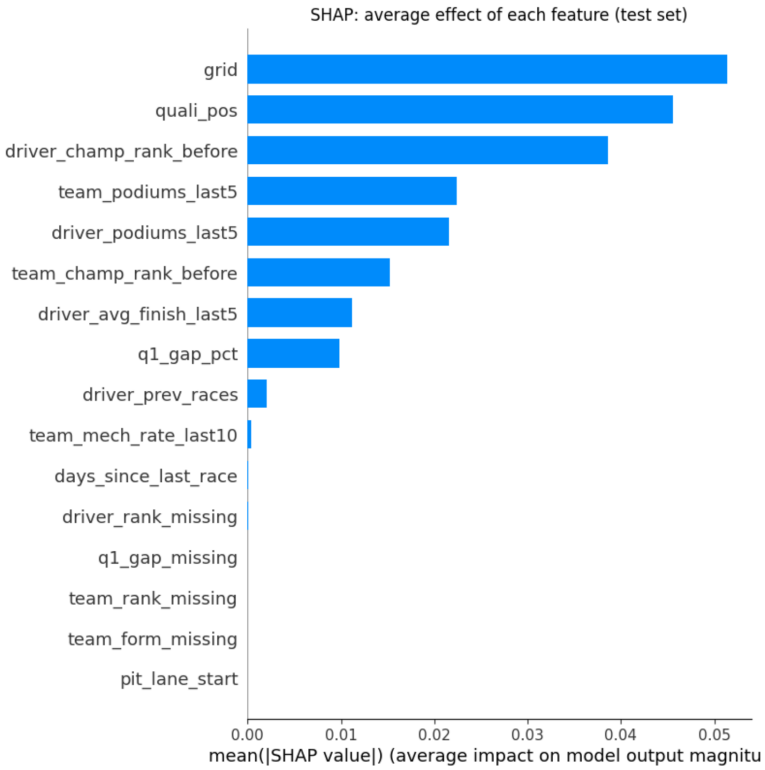


Figure 19. Mean absolute SHAP value per feature (test set).

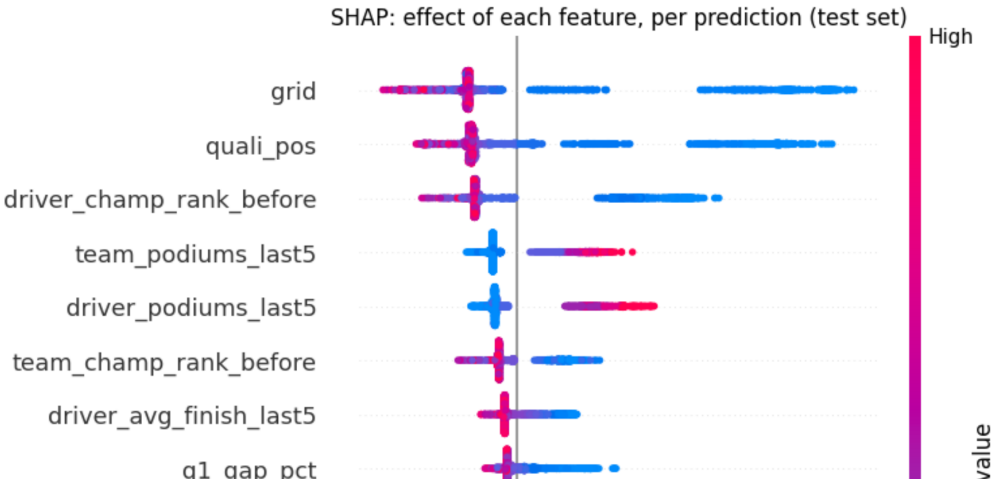


Figure 20. SHAP beeswarm for the top features (blue = low value, red = high value).

The top three match permutation importance. Recent podium counts rank higher with SHAP: for redundant features, shuffling one hurts little while SHAP shares credit between them. Low grid, qualifying position and championship rank push the predicted probability up, as do many recent podiums.

7.3 Local: one prediction

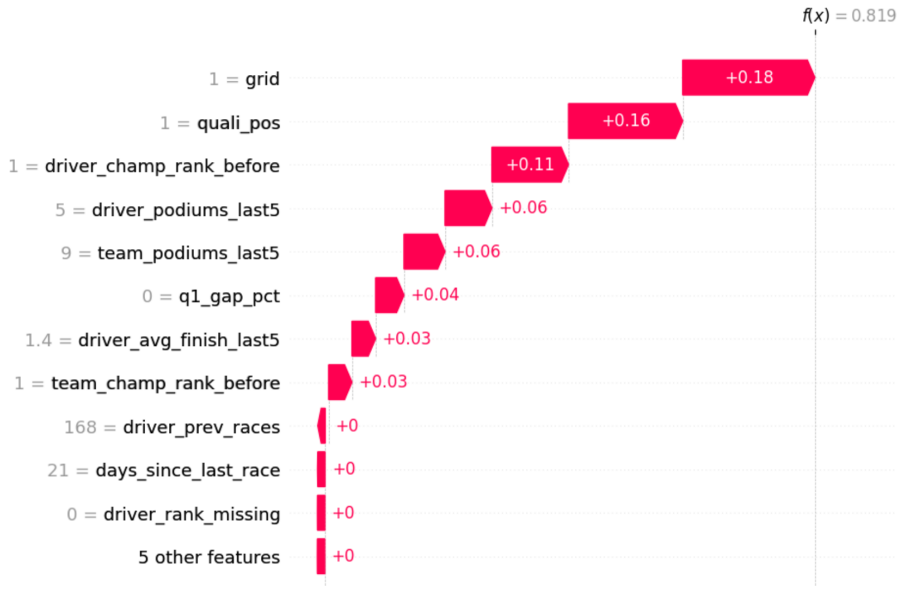


Figure 21. SHAP waterfall for the highest test prediction: Max Verstappen, 2023 Monaco Grand Prix (round 6), grid 1; predicted 0.819; actual podium.

Starting from pole (+0.18), qualifying first (+0.16) and leading the championship (+0.11) contributed most, followed by his and his team's recent podiums (+0.06 each).

8. Inference function

predict_podium(record) takes a dictionary of pre-race features and returns the podium probability, the decision at the validation threshold (0.31), and the list of missing inputs. It applies the training rules: missing podium counts → 0; missing grid or qualifying position → the other one; grid 0 → pit_lane_start = 1 and grid 21 (one place behind a 20-car grid); other missing values → training medians with the corresponding flags = 1. Note: training removed the 23 rows without a qualifying record, so copying grid into a missing qualifying position is used only for new records.

Input	Probability	Decision	Missing inputs
Complete test record	0.538 (identical to the model's direct prediction)	Podium	none
Same record with 4 values removed	0.455	Podium	quali_pos, q1_gap_pct, team_mech_rate_last10, driver_champ_rank_before

9. Summary of design decisions

Each decision below is justified in the section indicated; alternatives were rejected for the reason given.

Decision	Alternative rejected	Reason	Sec
Clean on copies; raw CSVs untouched	—	Brief requirement; enables before/after EDA	2
\N → NaN via na_values when loading	Replace text afterwards	Same result; numeric types detected automatically	2.1
Duplicates: own-grid-slot rule (53) + seeded random (32)	Keep best result; keep better grid; drop cases	Uses only pre-race grid; no second chance at a podium; exactly one row per driver and race	2.4
Fill 2 positions only after checking the slot is free	Leave empty	Verified consistency	2.5
Model 2003–2024, whole seasons	All years with filled qualifying	Complete qualifying; no invented values	2.6
Remove pre-race non-starters by status	Use laps = 0	Not real predictions; laps is post-race	2.6
Q1: grid as main definition + qualifying + three groups	One definition only	Brief asks to distinguish; groups show the penalty effect	3.1
Q2: grid groups + same-driver comparison	Grid groups only	Also removes driver quality as an explanation	3.2
Q3: mechanical only if a part is named; keep all teams	Count "Retired"; fixed minimum of starts	No evidence of failure; every team has ≥ 31 starts	3.3
No home or sprint feature; rank instead of points	Include them	Q2 found no effect; sprints absent in training; points system changed	4.2
Features on full history, cut to 2003+ afterwards	Compute on 2003+ only	Early 2003 rows need earlier races	4.2
Teams reset after >4 years away; drivers keep history + days_since_last_race	Reset drivers after 1–2 years	Returning teams are new organisations; a driver is the same person	4.2
Q1-only qualifying gap	Best of Q1–Q3	Same session for all drivers	4.2
Remove 23 rows without qualifying	Copy grid into quali_pos	No substitute values in training	4.3
0 / training median + missing flags	Drop 330 rows	Keeps rookies and new teams; same rule at inference	4.3
Train 2003–18 / val 2019–21 / test 2022–24	Validation 2020–21	~180 instead of 117 validation podiums	4.4
PR-AUC primary, ROC-AUC secondary; threshold on validation F1	Accuracy; threshold 0.5	Rare positives; test never used for choices	5.2
Final model: random forest, depth 4	Logistic regression, neural network	Best validation PR-AUC and best on test	5.3

10. Limitations and conclusion

- **Qualifying-coverage drift**: no qualifying before 1994 and patchy coverage until 2002 restrict modeling to 2003–2024.
- **Era sensitivity**: regulation changes (2014, 2022) and the 2010 points change shift patterns; test performance drops slightly in the 2022 era.
- **Redundant features**: form and championship groups overlap, so single-group ablation understates their joint value.
- **Small samples**: per-circuit Q1 rates, home races in Q2, and single-season teams in Q3.
- **Definitions**: renamed teams reset history; nationality may differ from home; the mechanical-status mapping and the unresolved shared drives (32, random) are judgement calls.
- **Non-causal explanations**: XAI reflects model reliance only.

Using only pre-race information, a depth-4 random forest reaches a test PR-AUC of 0.721 and ROC-AUC of 0.940 on 2022–2024, about 4.8 times the PR-AUC of a random model. Starting position is the dominant signal, as both the analysis (Q1) and the model explanations show, while recent form, championship position and qualifying speed add smaller, partly overlapping information.

Appendix A. Effect of the preprocessing on the data

Section 4.3 says which values are filled and that the features are standardised. This appendix shows what those two steps do to the data itself: their effect on each feature's distribution, range and values. The "before" column is the feature before filling. The filled rows are exactly those marked by the missing flags, and rookies by driver_prev_races = 0, so they can be set back to empty and compared with the filled version. All numbers below are the training seasons only (2003–2018, 6,362 rows), the only part of the data used to learn the medians and the scaler; the counts in Section 4.3 cover all seasons, so they are larger.

Feature	Filled	Range before	Mean: before → filled	SD: before → filled	Range after scaling
grid	0	1.00 to 24.00	11.207 → 11.207	6.245 → 6.245	−1.63 to 2.05
quali_pos	0	1.00 to 24.00	11.206 → 11.206	6.245 → 6.245	−1.63 to 2.05
q1_gap_pct	94	0.00 to 33.29	1.814 → 1.807	1.858 → 1.845	−0.98 to 17.06
driver_avg_finish_last5	72	1.00 to 24.00	11.150 → 11.153	4.533 → 4.507	−2.25 to 2.85
days_since_last_race	72	7 to 3,584	23.809 → 23.698	82.377 → 81.916	−0.20 to 43.47
team_mech_rate_last10	41	0.00 to 1.00	0.122 → 0.122	0.113 → 0.112	−1.09 to 7.81
driver_champ_rank_before	151	1.00 to 27.00	11.323 → 11.315	6.510 → 6.432	−1.60 to 2.44
team_champ_rank_before	65	1.00 to 12.00	5.840 → 5.841	3.120 → 3.104	−1.56 to 1.98

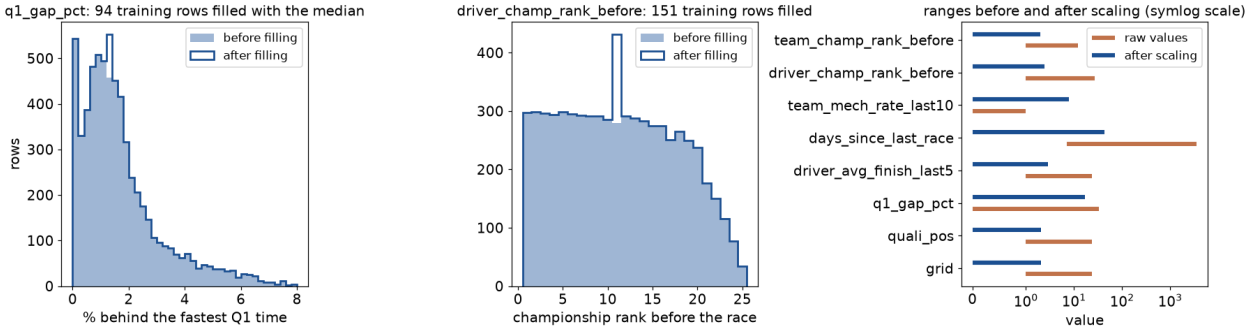


Figure 22. What the two preprocessing steps change. Left and middle: the distribution of a filled feature before and after median filling. Right: the range of each feature before and after scaling, on a symmetric log scale.

Filling moves a feature's mean by at most 0.11 (days_since_last_race) and its standard deviation by at most 0.5, because only 41 to 151 of the 6,362 training rows are filled per feature. What it does add is a spike at the median, visible in both histograms, which is why every filled value also carries a flag: the model can separate "filled" from "really average". Scaling changes no shape and no ordering. It maps each feature onto a common scale (mean 0, standard deviation 1 on the training seasons), so ranges as different as 0–1 for team_mech_rate_last10 and 7–3,584 for days_since_last_race become comparable for logistic regression and the neural network, which both add up weighted features. The skew survives: days_since_last_race still reaches 43 standard deviations above its mean, which is one reason the random forest, whose splits do not depend on scale, handles these features more comfortably. The code and the output are in section 11 of the notebook.

Appendix B. Local explanation with LIME

Section 7.3 explains one prediction with a SHAP waterfall. LIME answers the same question in a different way, and the checklist asks for it by name. SHAP splits the prediction into contributions that add up to the model's output. LIME instead samples points around this one driver-race, weights them by how close they are to it, and fits a small linear model to the random forest's answers in that neighbourhood; its weights describe the model near this row, not overall. We run it on the same row as the waterfall: Max Verstappen, 2023 Monaco Grand Prix (round 6), grid 1, the highest predicted probability on the test seasons (0.819, and a real podium).

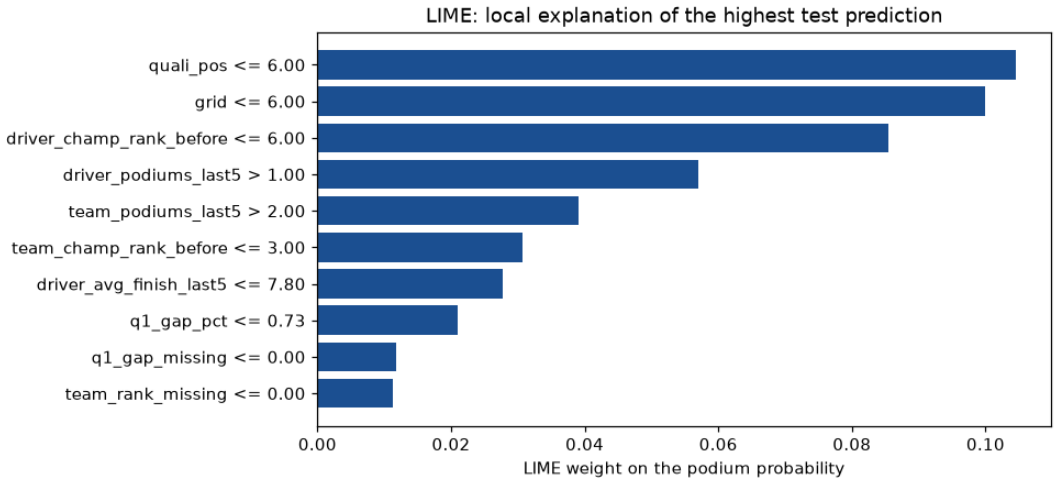


Figure 23. LIME weights for the same prediction as Figure 21. Each bar is a condition on a feature and how much it moves the predicted podium probability in the neighbourhood of this row.

LIME puts the same facts on top as SHAP: qualifying in the first six (+0.10), starting in the first six (+0.10) and a championship rank in the top six (+0.09), then the driver's and the team's recent podiums (+0.06 and +0.04) and a fast Q1 (+0.02). The two methods agree on which pre-race facts the model is using for this driver, and on their order, which is the useful part of the comparison.

The surrogate itself is only a rough fit: it predicts 0.473 where the random forest predicts 0.819, with an R-squared of 0.763 in that neighbourhood. A straight line through a sampled neighbourhood cannot reproduce the level of a forest built from threshold splits, so we read the direction and the ranking of the LIME weights and rely on the SHAP waterfall for the exact decomposition of the prediction. As everywhere in Section 7, these are statements about what the model relies on, not about what causes a podium finish: grid and qualifying position are informative here because they summarise car and driver pace, which is also what produces the result on Sunday. The code and the output are in section 13 of the notebook.